

EE4C10

Analog Circuit Design Fundamentals

Week Q1.5
September 30th, 2025
Fabio Sebastiano

Today's goal

- Get more familiar with current mirrors and differential amplifiers
 1. Practical with LTspice
 2. Practice differential pair calculations
 - Using CM and DM simplifications

LTspice practical

Exercise 1 – LTspice

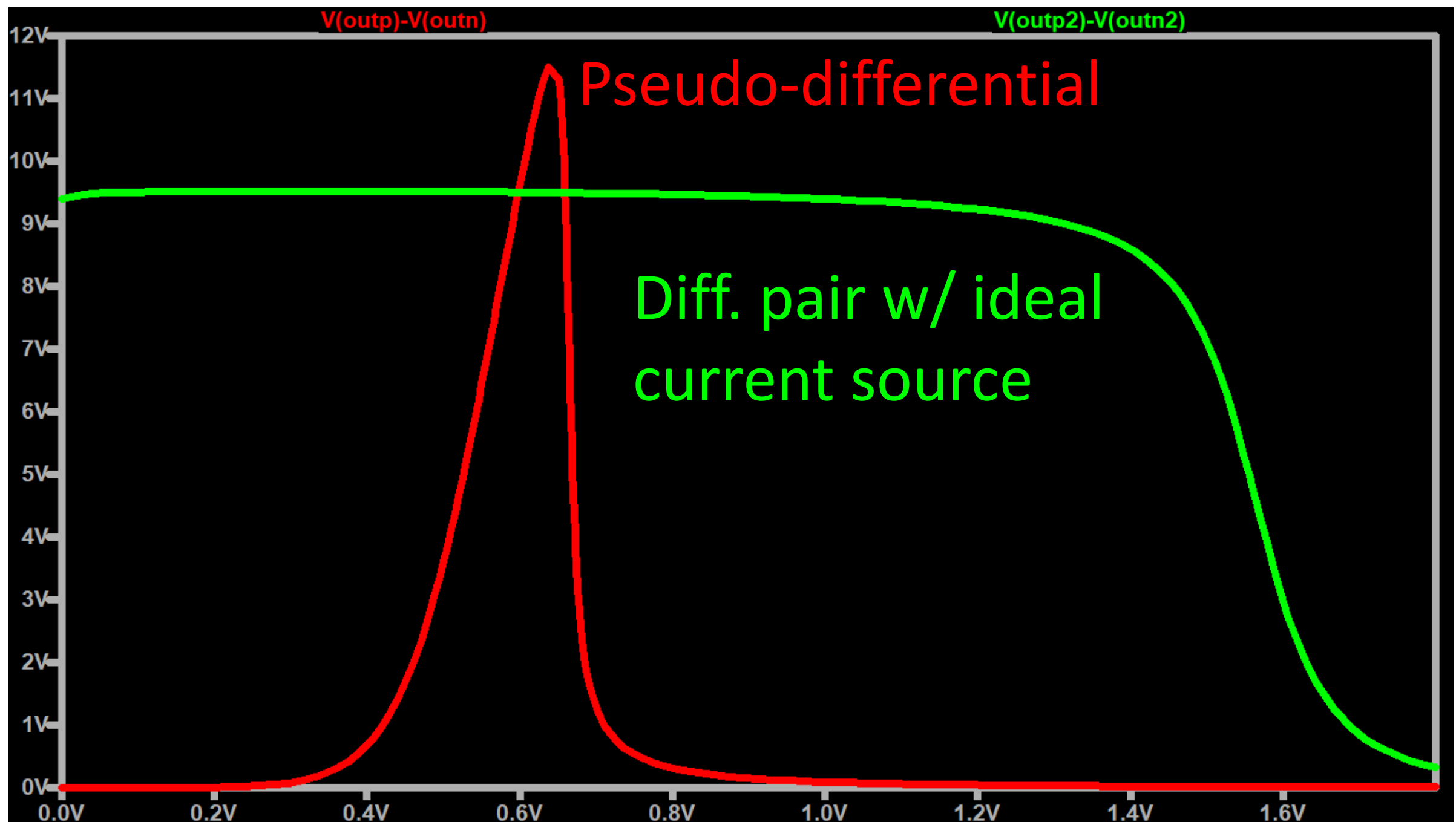
Use the file on Brightspace: **Week4_Ex1.asc**. It contains the schematic of a pseudo-differential pair and of a differential pair.

- A. Plot the small-signal differential gain while sweeping the input CM from 0 V to 1.8 V. Explain the behaviour.

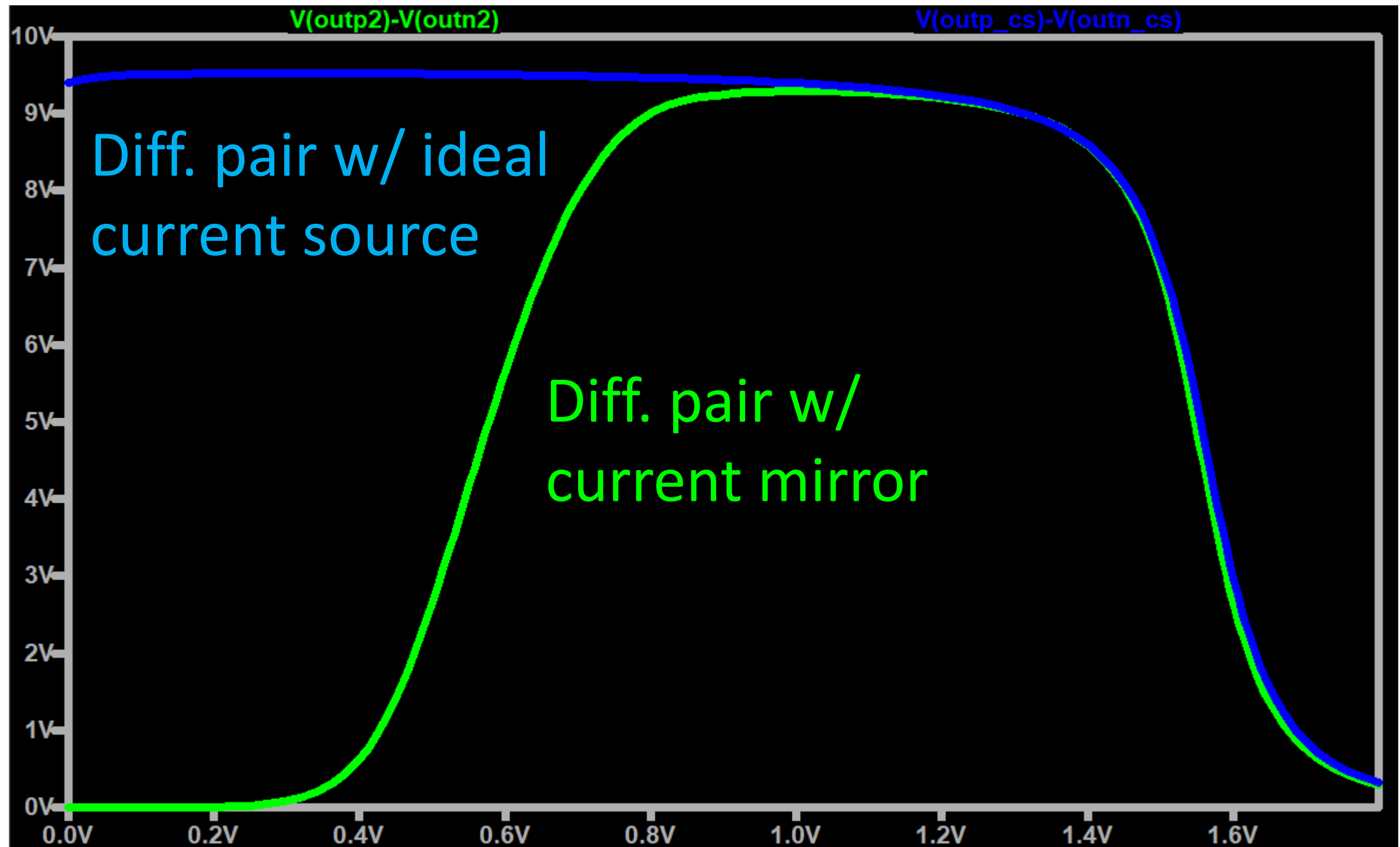
You can use `.step VCM 0 1.8 10m`
 `.ac list 1`

- B. Copy the diff. pair circuit and replace the ideal current source with a current mirror. Repeat point A for the new circuit. Explain the differences of the new circuit.
- C. Simulate the CM gain for the 3 circuits. What is the CMRR?
- D. Add a mismatch of 1% in the load resistors. Plot A_{CM-DM} for the 3 circuits.

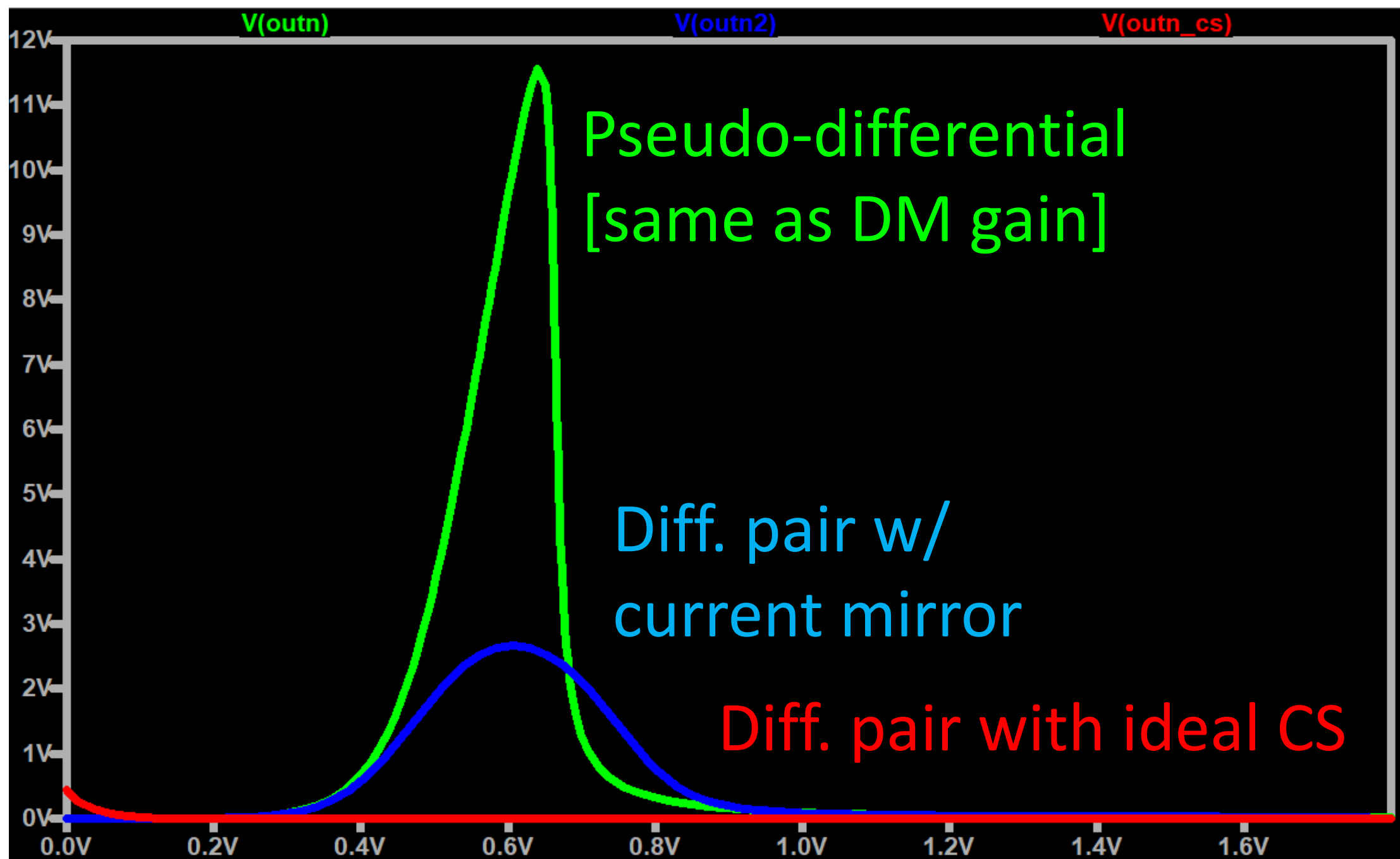
Results – Differential gain



Results – Differential gain

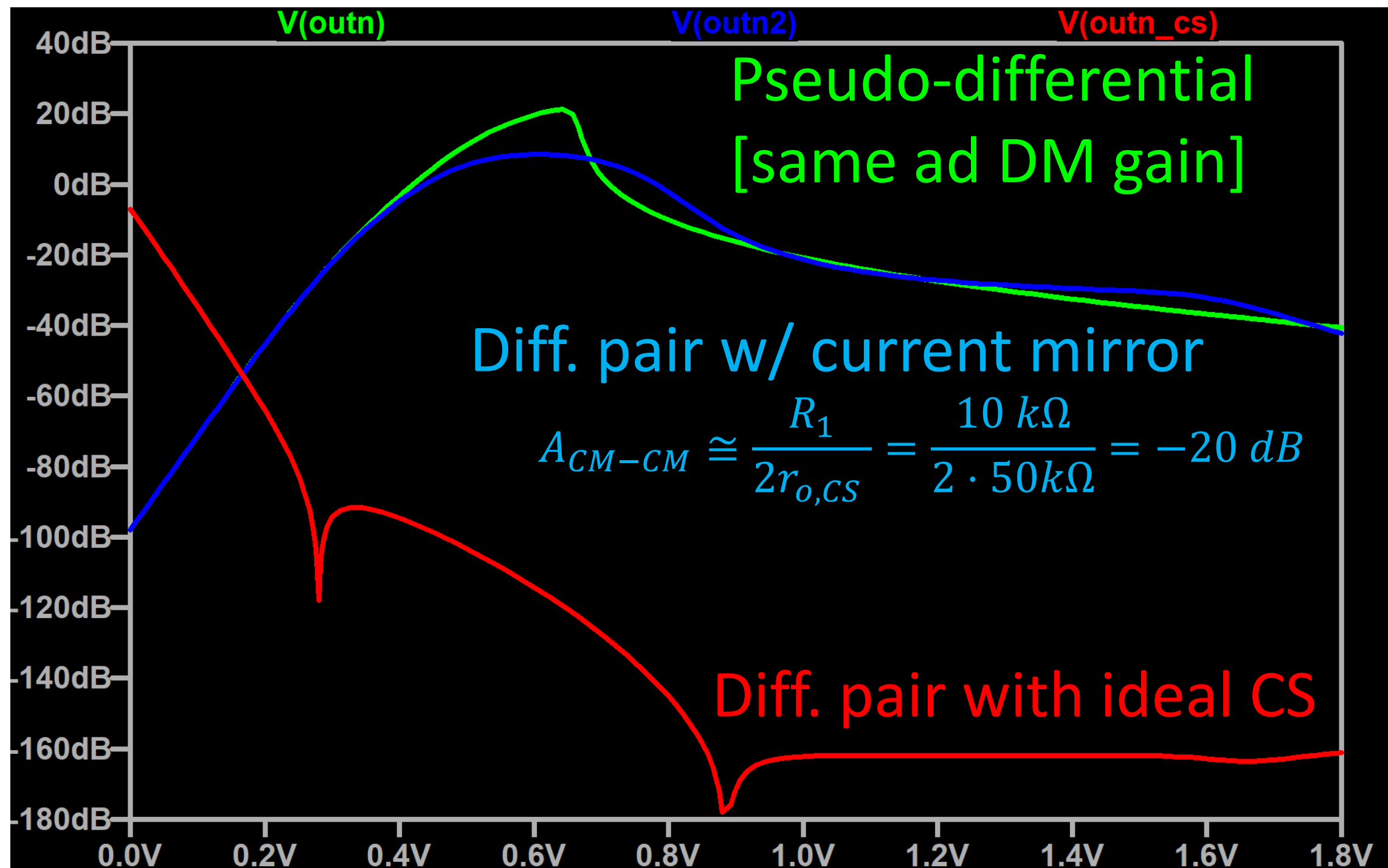


Results – CM gain



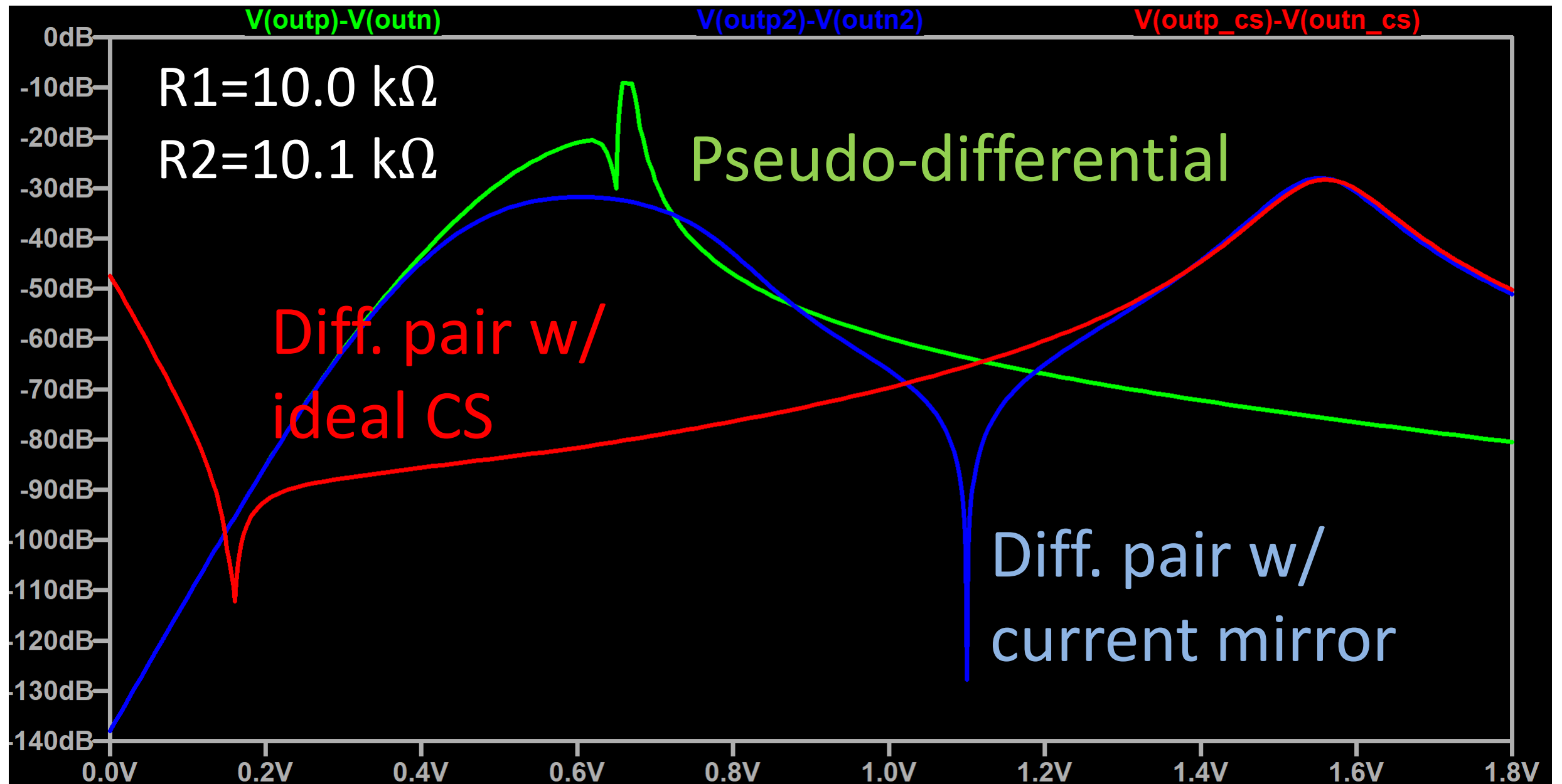
Note: CMRR = ∞ since $A_{CM-DM} = 0$

Results – CM gain on dB scale



Note: CMRR = ∞ since $A_{CM-DM} = 0$

Results – CM-DM gain



Exercise 2 – LTspice

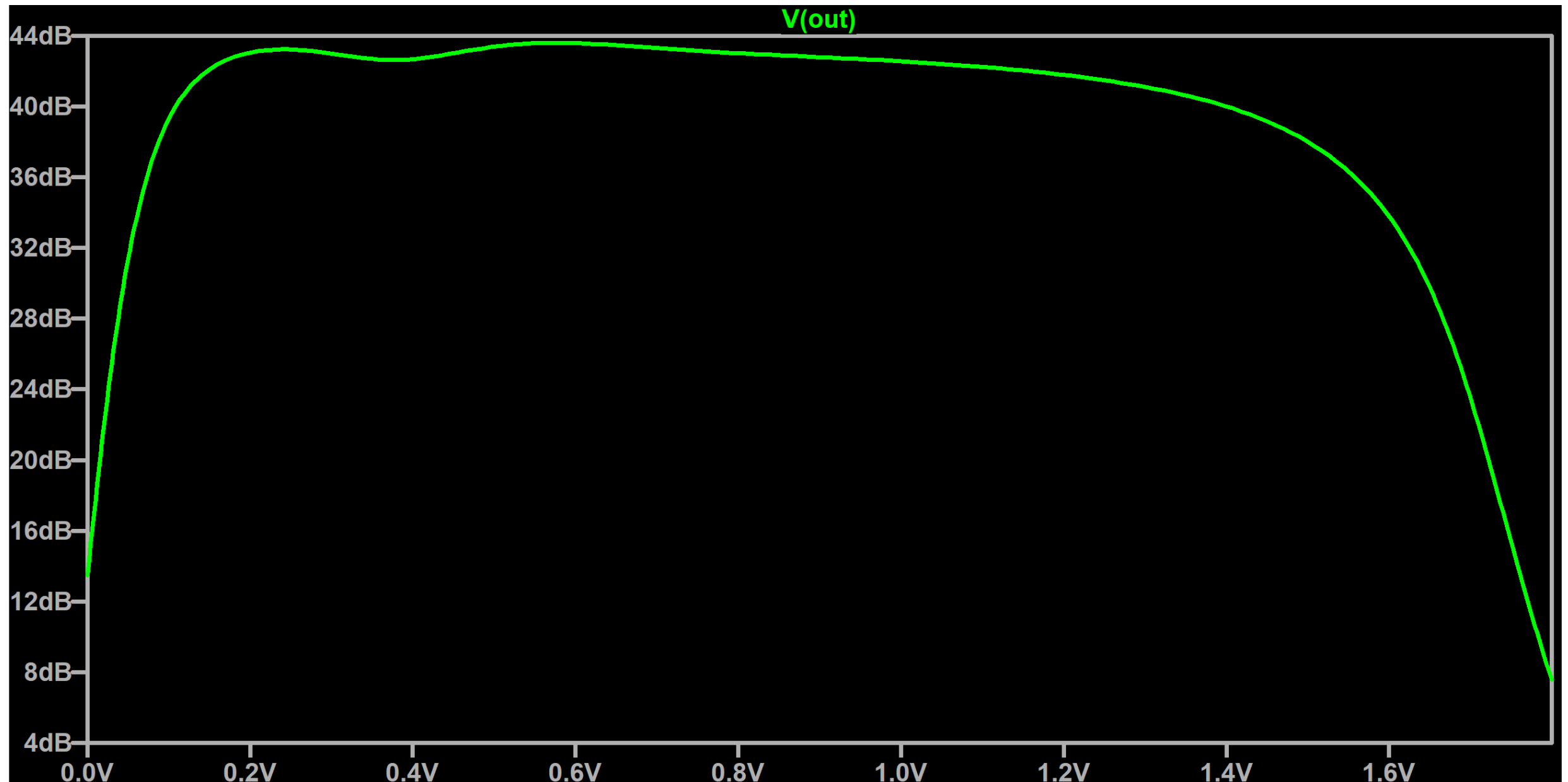
Use the file on Brightspace: **Week4_Ex2.asc**. It contains the schematic of a 5-transistor OTA.

- A. Find the small-signal DM gain both by AC analysis and by sweeping the DM input.
- B. Plot the small-signal differential gain while sweeping the input CM from 0 V to 1.8 V. Explain the behaviour.

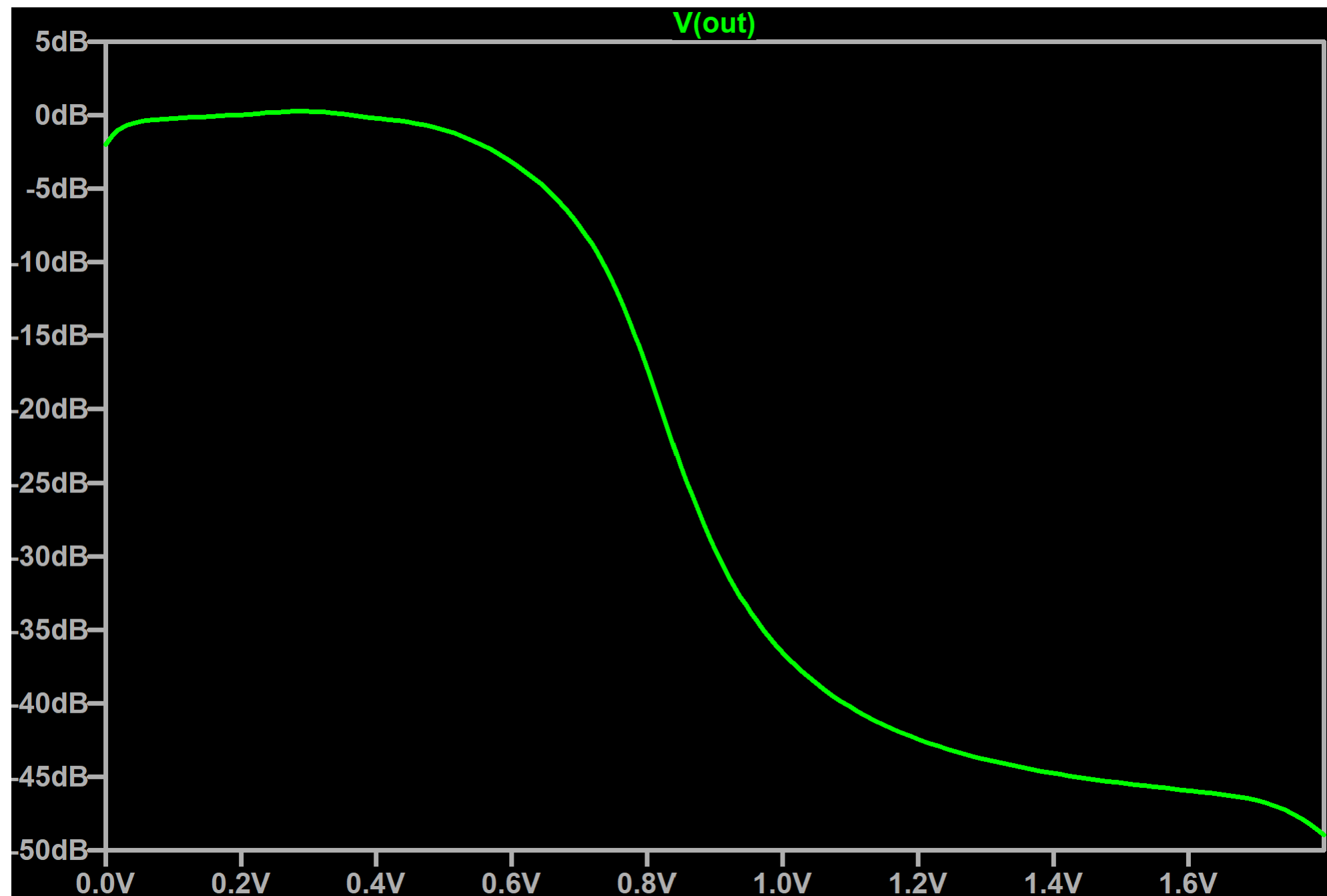
You can use `.step VCM 0 1.8 10m`
 `.ac list 1`

- C. Simulate the CM gain. What is the CMRR?
- D. Connect the output to the negative input, i.e. connect the Ota as a buffer. Add an AC excitation on the positive input. Plot the gain as a function of the DC value on the positive input.

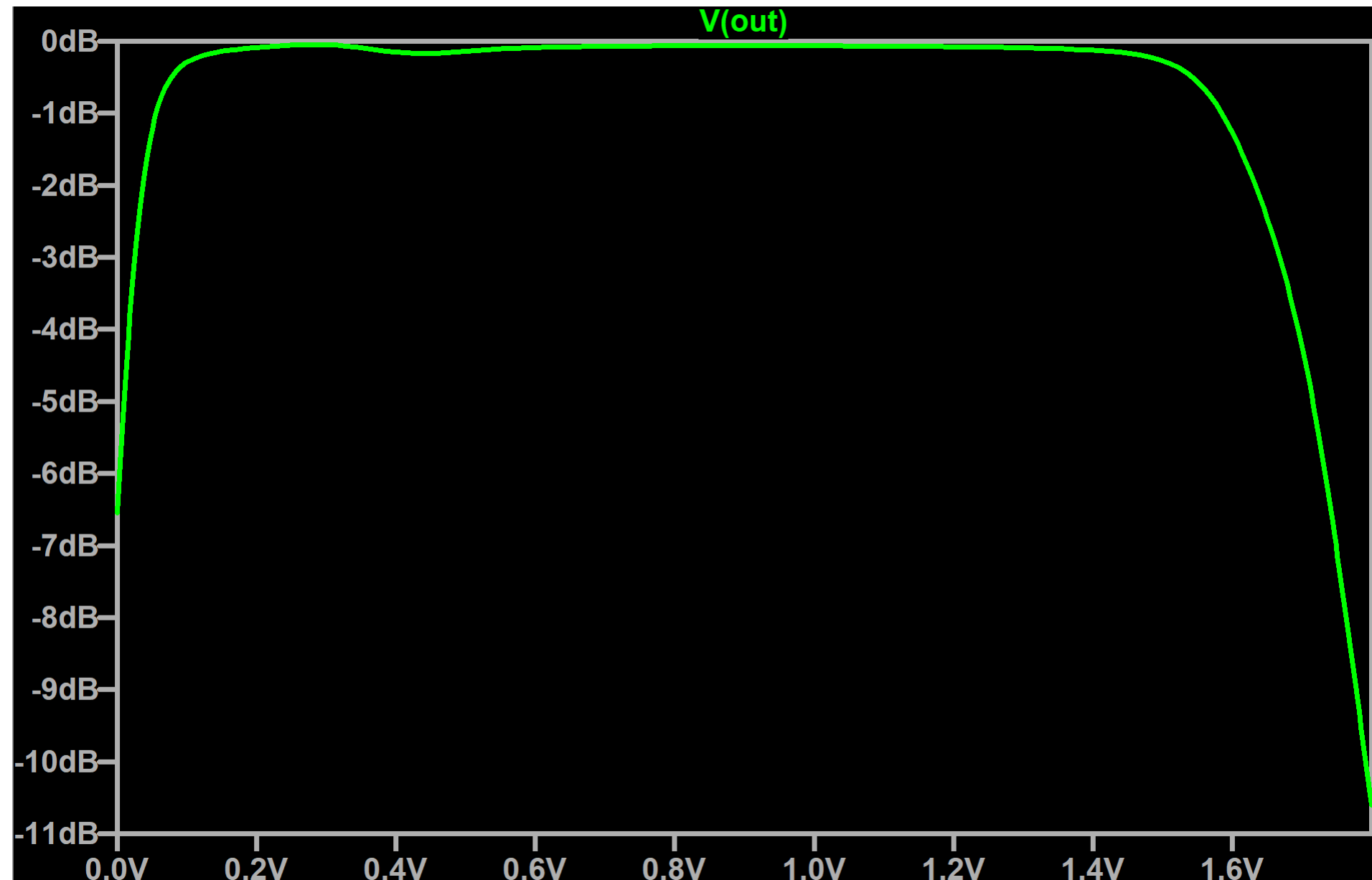
Result – DM gain vs. input CM



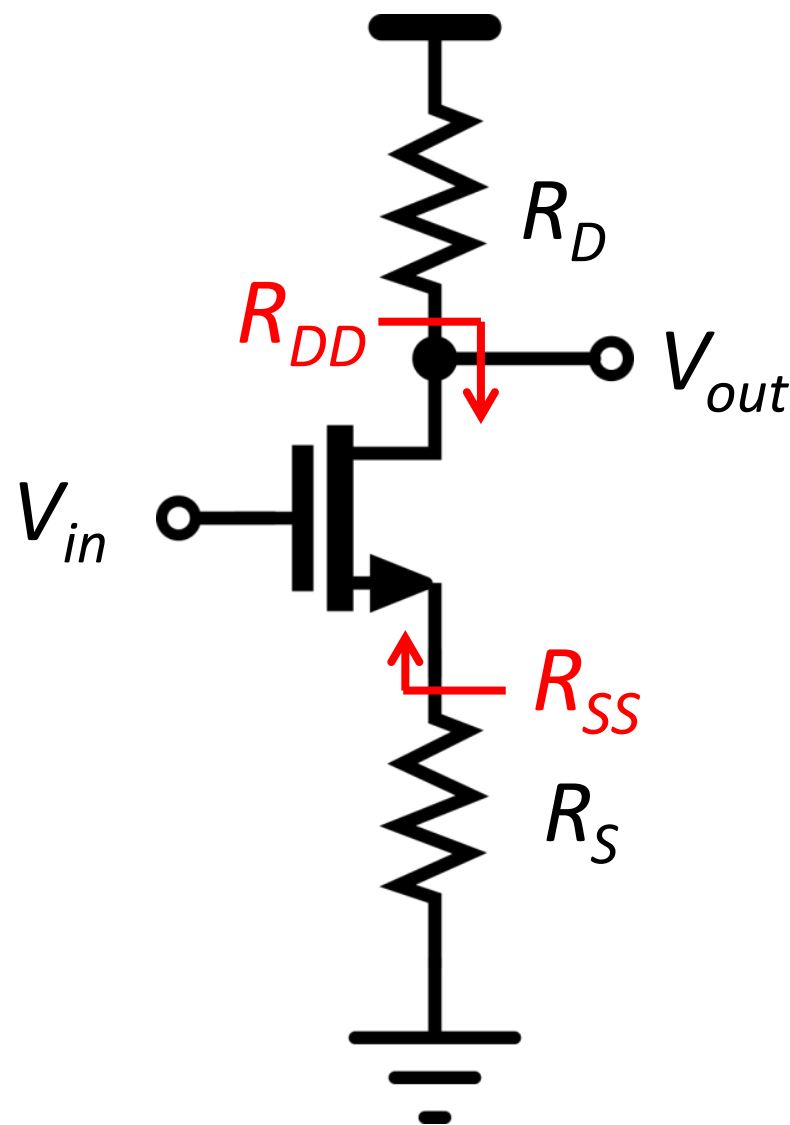
Results – CM gain vs. input CM



Result – Buffer gain



Quick recap



- Resistance from drain

$$R_{DD} = r_o[1 + (g_m + g_{mb})R_S] + R_S$$

$$\cong r_o + R_S + g_m r_o R_S$$

$$R_{SS} = \frac{r_o + R_D}{1 + g_m r_o} \cong \frac{1}{g_m}$$

- Norton-equivalent from drain (w/o R_D)

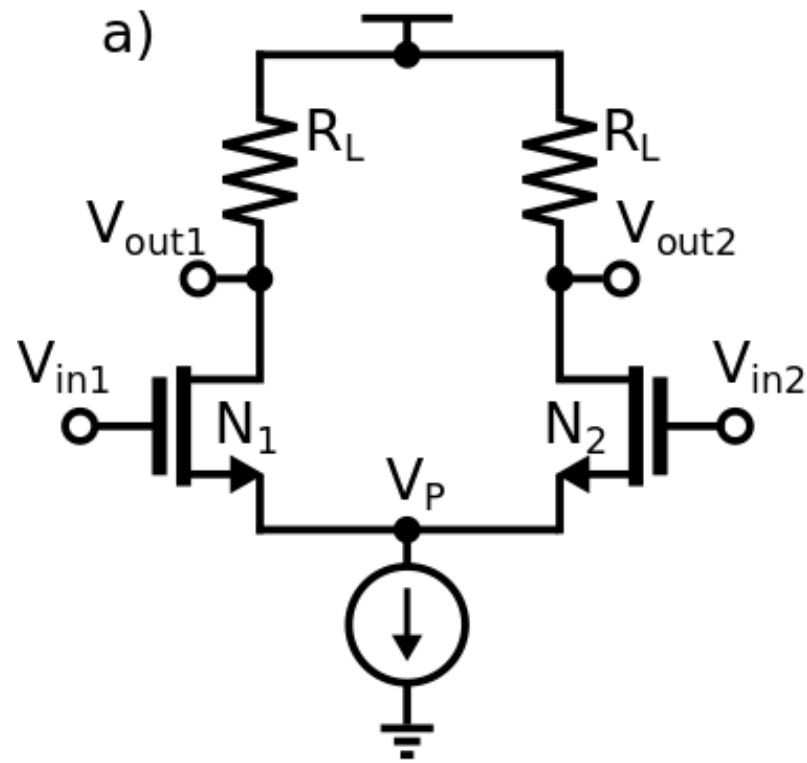
$$G_m = \frac{g_m r_o}{r_o[1 + (g_m + g_{mb})R_S] + R_S}$$

$$\cong \frac{g_m}{1 + g_m R_S}$$

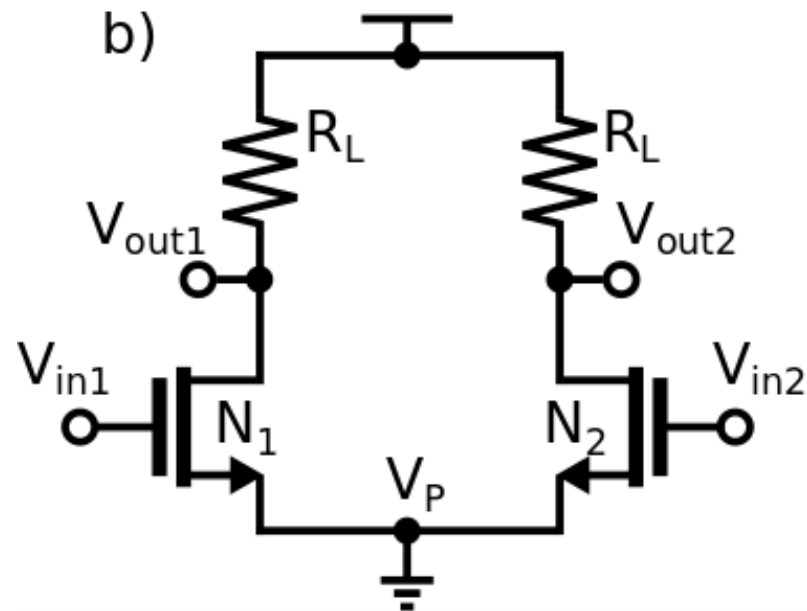
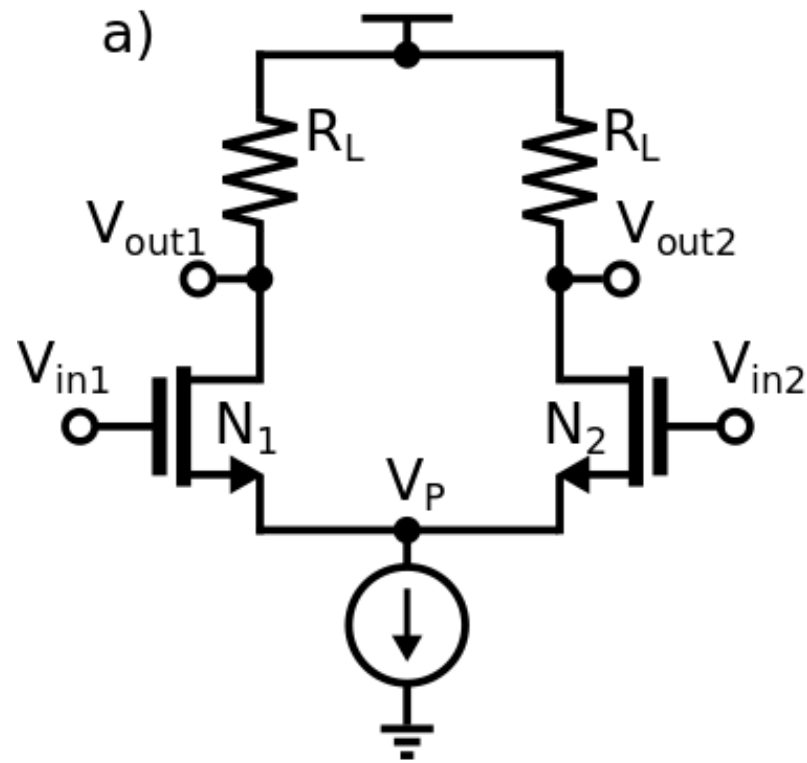
- Approximations assume

- $r_o \gg R_D, R_S$
- Neglect g_{mb}

Compute diff. gain and CM gain



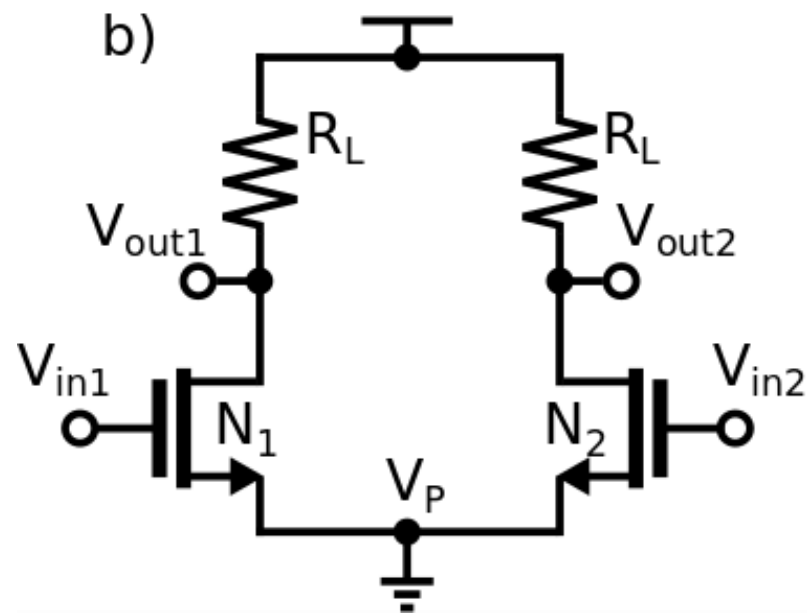
Compute diff. gain and CM gain



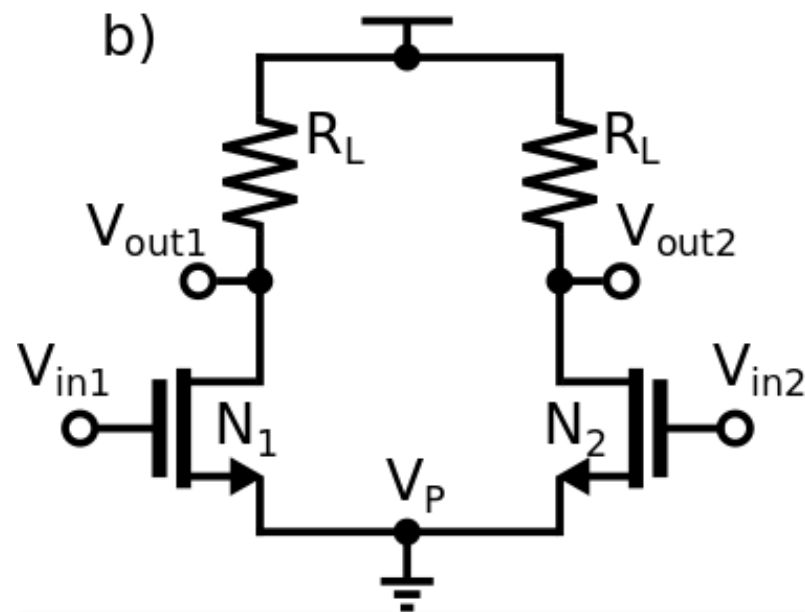
Compute diff. gain and CM gain

- $A_{DM-DM} = -g_m(r_o // R_L)$
- $A_{CM-CM} = -\frac{g_m r_o R_L / 2}{\frac{r_o}{2}(1 + 2g_m R_S) + R_S + \frac{R_L}{2}} \approx -\frac{R_L}{2R_S}$
- $A_{DM-DM} = -g_m(r_o // R_L)$
- $A_{CM-CM} = -g_m(r_o // R_L)$

Power supply rejection



Power supply rejection

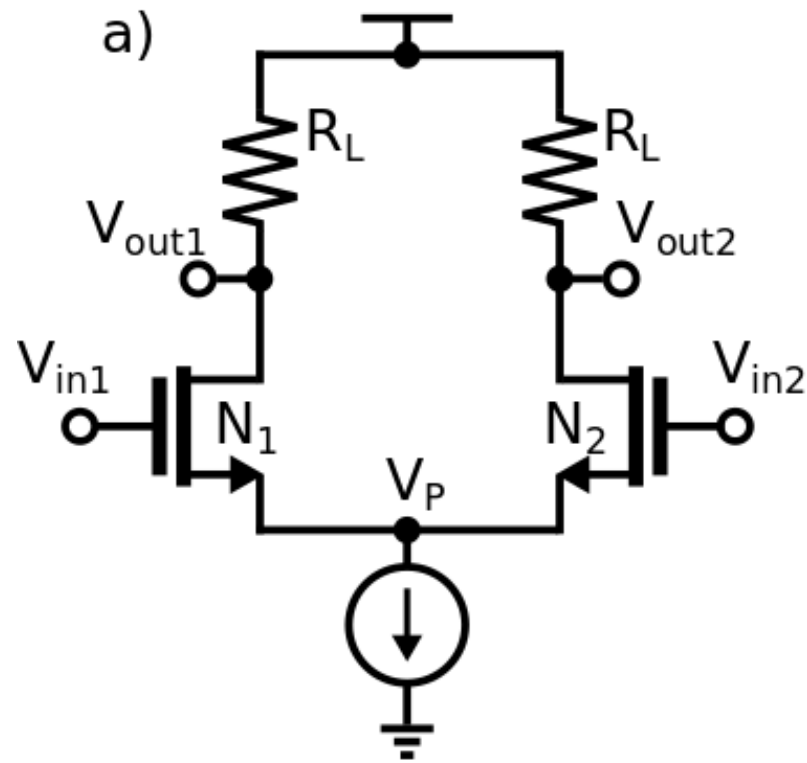


- $v_{o1} = v_{o2} = \frac{r_o}{r_o + R_L} v_{dd} \approx v_{dd}$
- If $R_{L1} \neq R_{L2}$

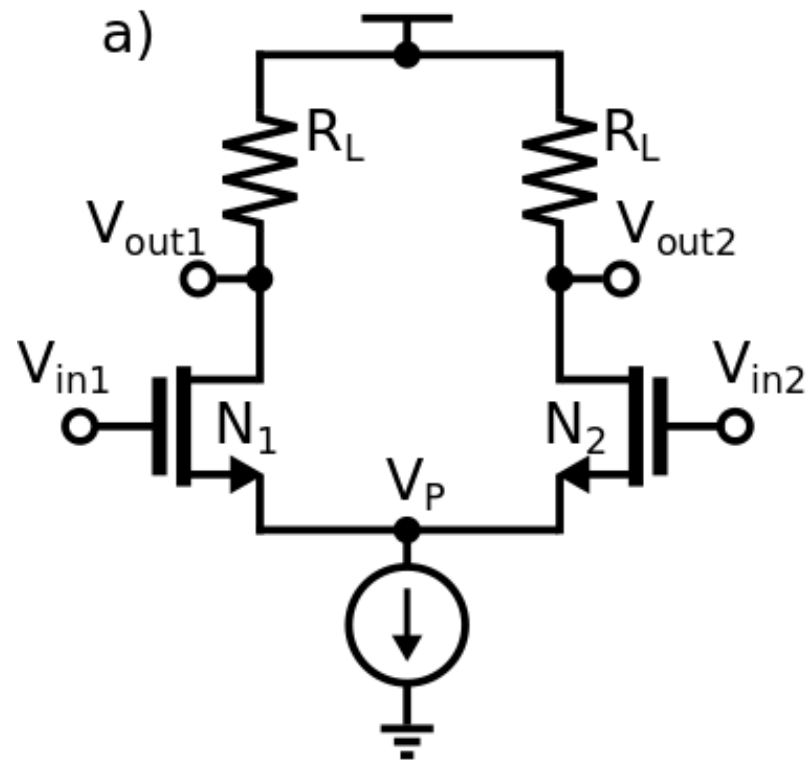
$$v_{o1} - v_{o2} \approx \frac{r_o \Delta R}{(R_L + r_o)^2} v_{dd}$$

$$\approx \frac{\Delta R}{r_o} v_{dd}$$

Power Supply rejection



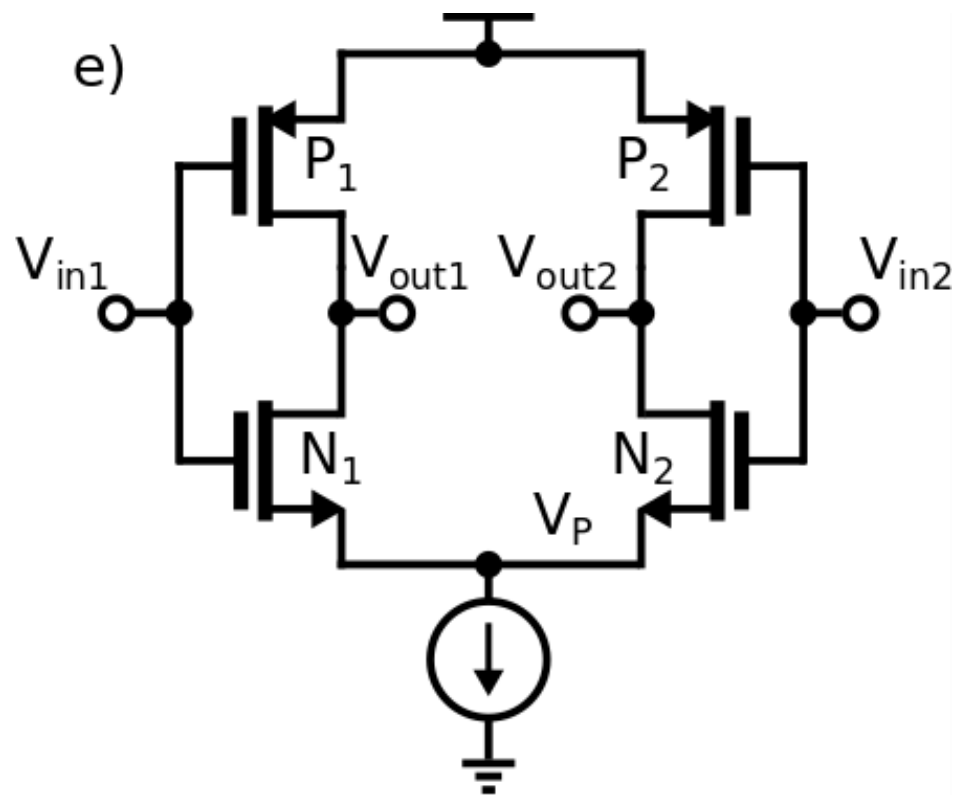
Power Supply rejection



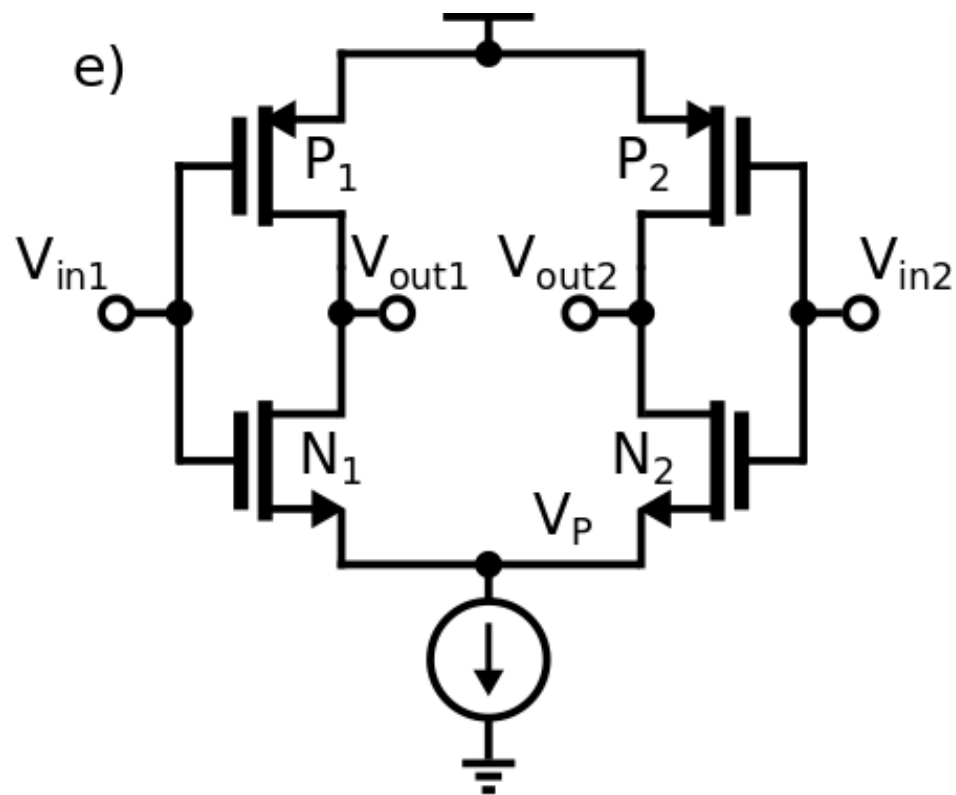
- If $R_{L1} \neq R_{L2}$

$$\frac{v_{o1} - v_{o2}}{v_{dd}} \approx \frac{\Delta R}{r_o + 2R_S}$$

Compute diff. gain and CM gain

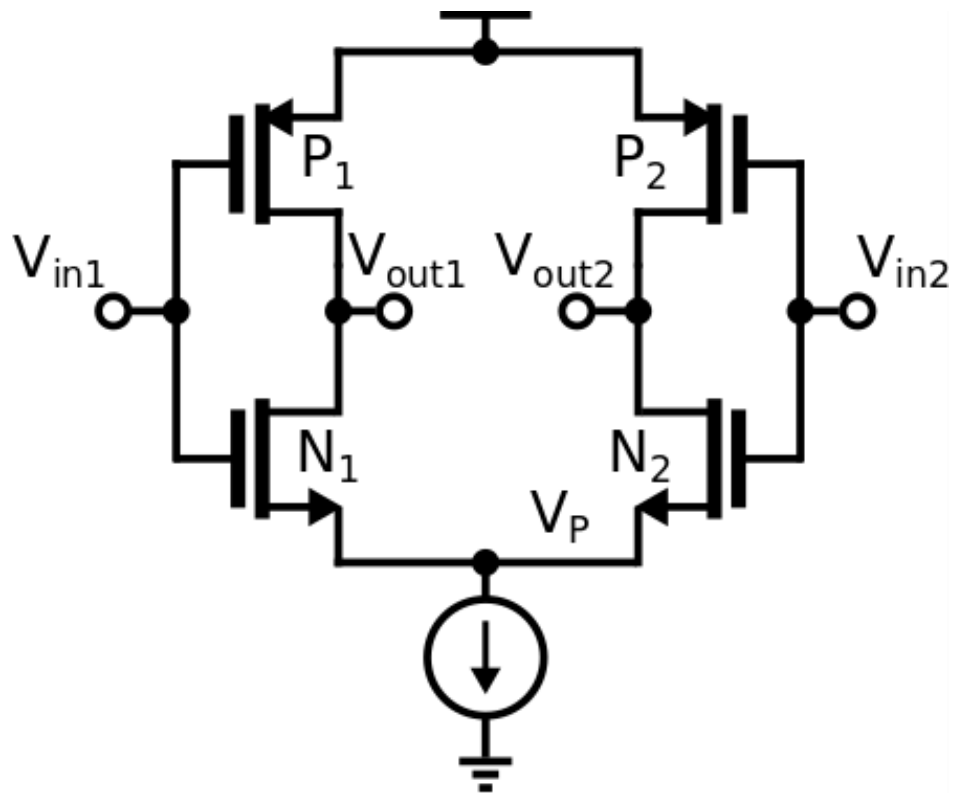


Compute diff. gain and CM gain

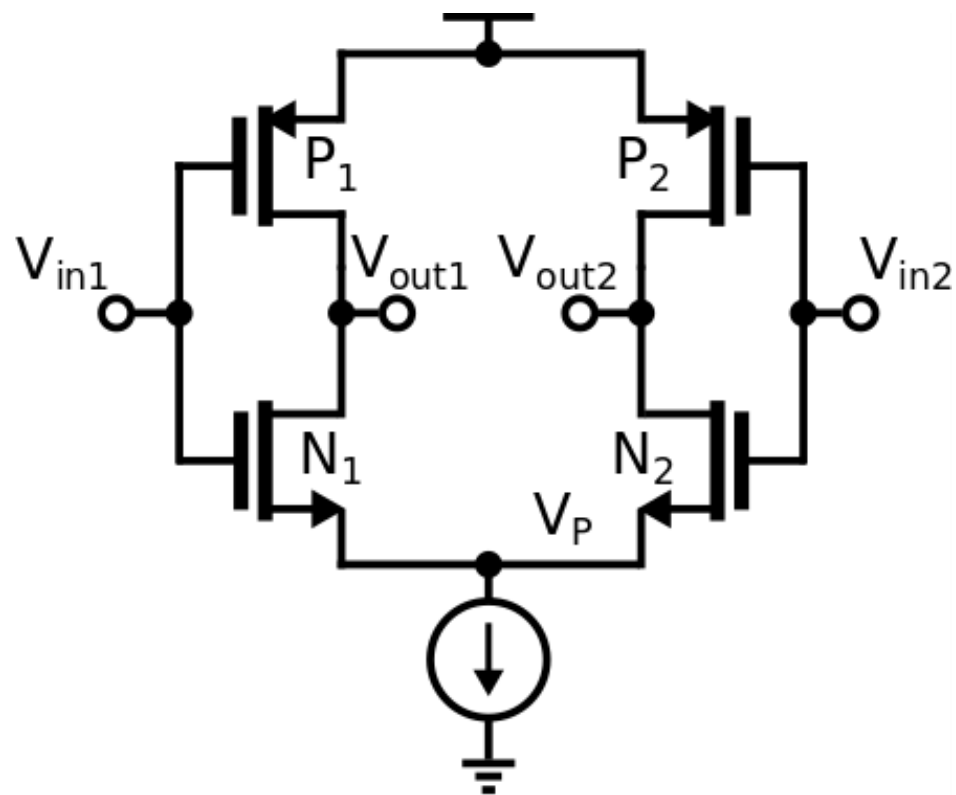


- $A_{DM-DM} = -(g_{mn} + g_{mp})(r_{op} // r_{on})$
- $A_{CM-CM} \approx -g_{mp}r_{op}$

Compute supply rejection

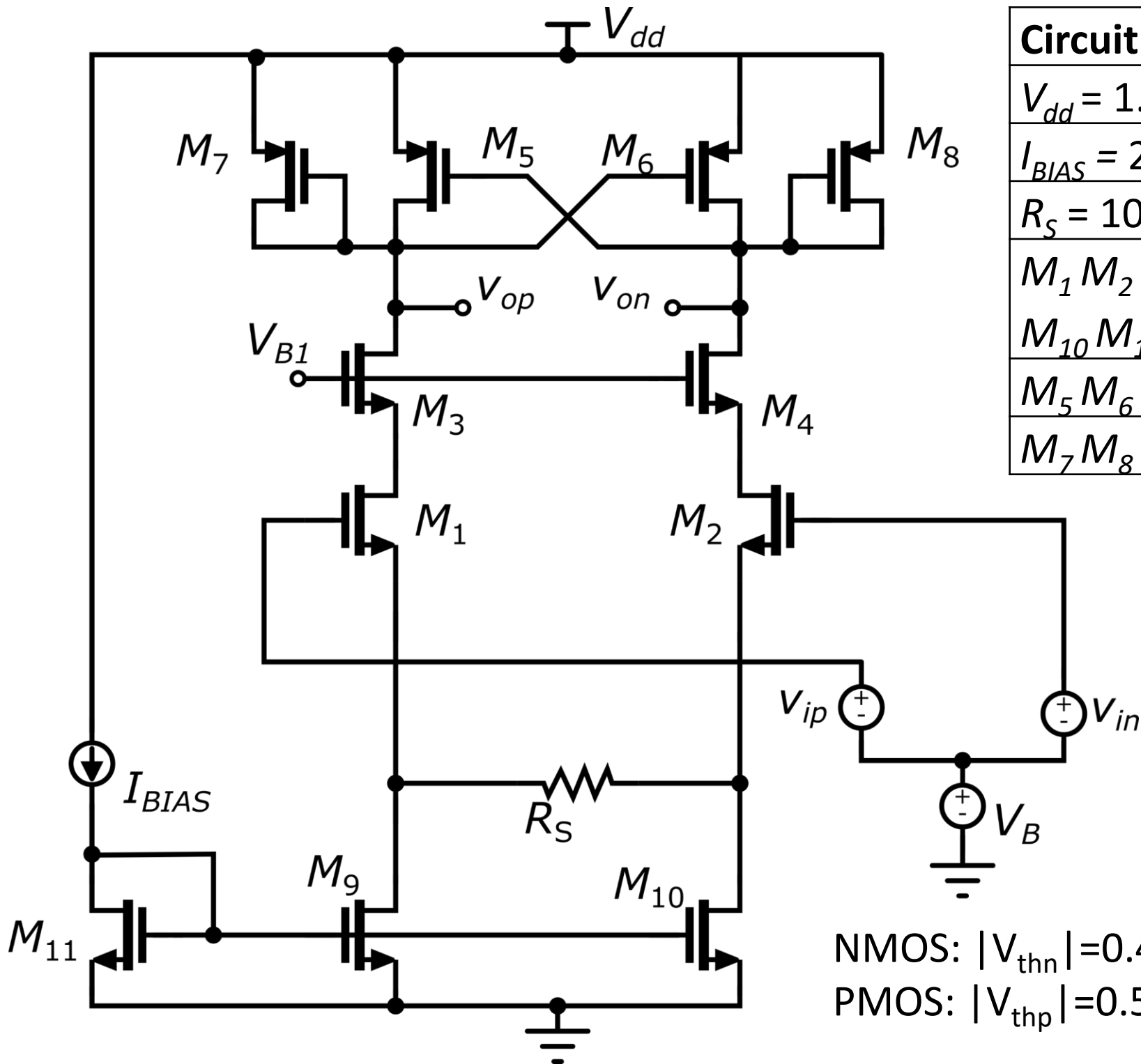


Compute supply rejection



- $v_{o1} = v_{o2} \approx g_{mp} r_{op} v_{dd}$

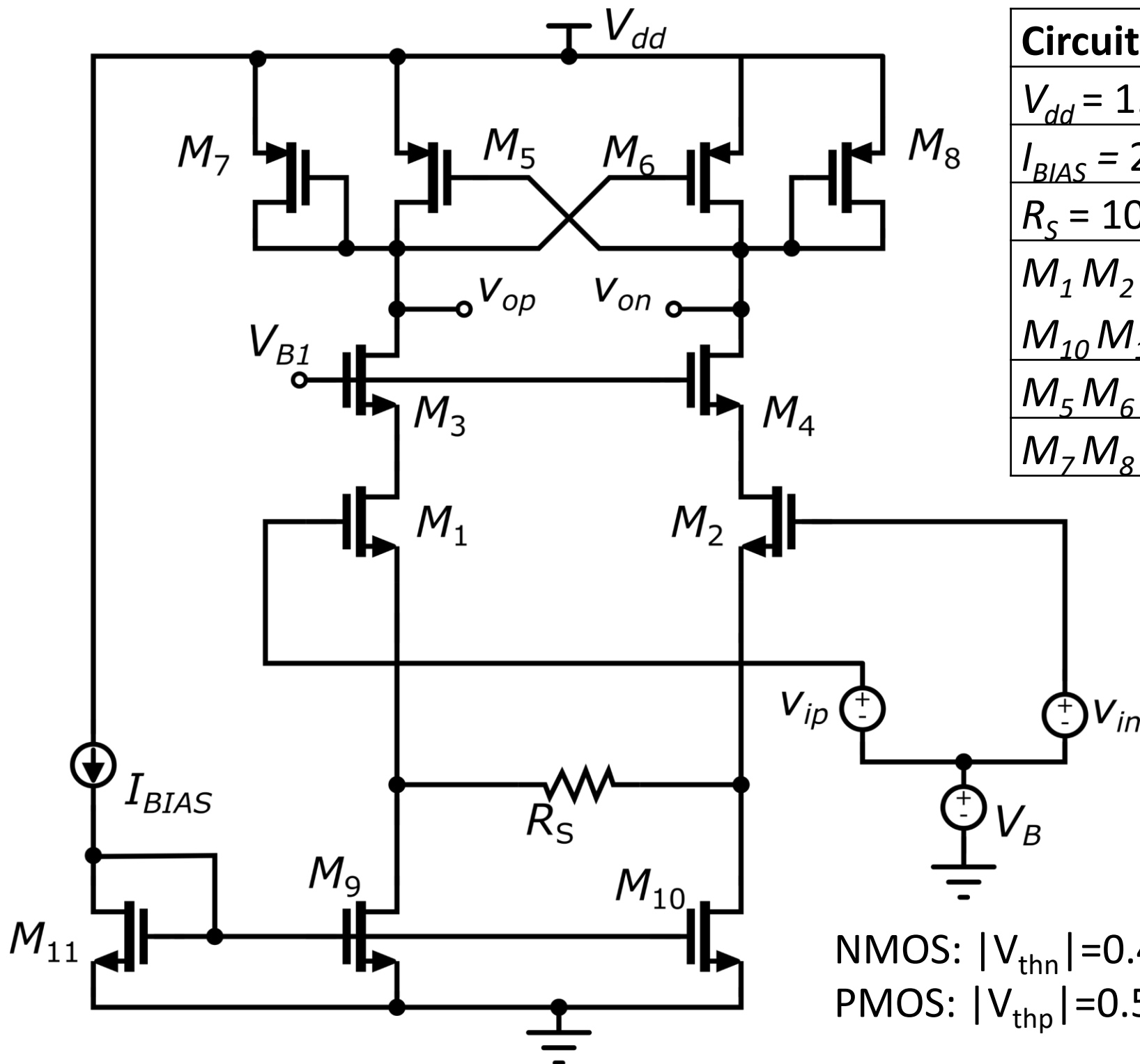
Bias point?



Circuit parameters	
$V_{dd} = 1.8 \text{ V}$	$V_B = 0.9 \text{ V}$
$I_{BIAS} = 20 \text{ }\mu\text{A}$	$V_{B1} = 1 \text{ V}$
$R_S = 100 \text{ k}\Omega$	
$M_1 M_2 M_3 M_4 M_9$ $M_{10} M_{11}$	$W = 1 \text{ }\mu\text{m}, L = 0.3 \text{ }\mu\text{m}$
$M_5 M_6$	$W = 1 \text{ }\mu\text{m}, L = 0.3 \text{ }\mu\text{m}$
$M_7 M_8$	$W = 2 \text{ }\mu\text{m}, L = 0.3 \text{ }\mu\text{m}$

NMOS: $|V_{thn}|=0.4\text{ V}$; $\mu_n C_{ox}=300\text{ }\mu\text{A/V}^2$; $\lambda_n=0\text{ V}^{-1}$
 PMOS: $|V_{thp}|=0.5\text{ V}$; $\mu_p C_{ox}=100\text{ }\mu\text{A/V}^2$; $\lambda_p=0\text{ V}^{-1}$

Bias point?

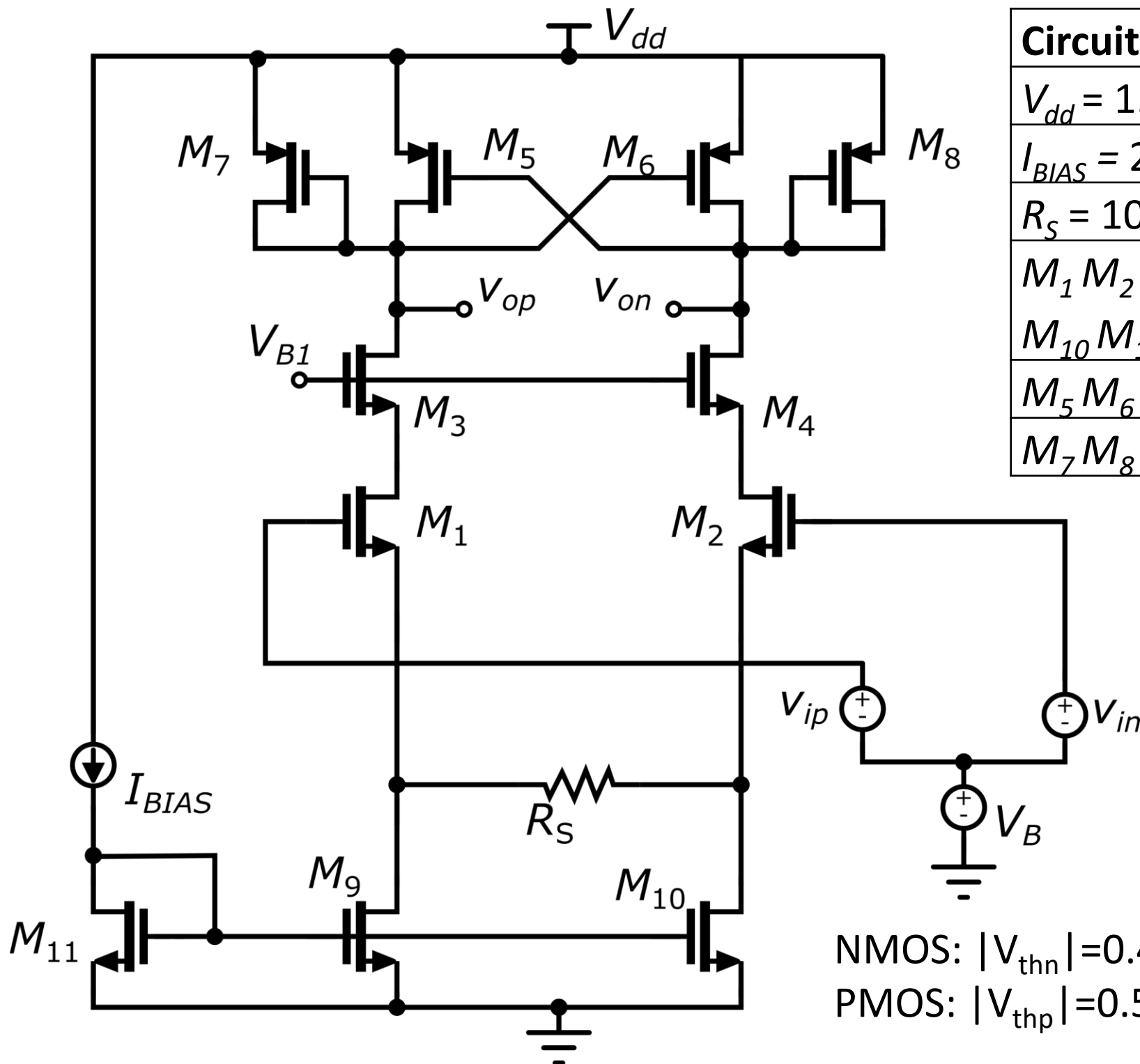


Circuit parameters

$V_{dd} = 1.8 \text{ V}$	$V_B = 0.9 \text{ V}$
$I_{BIAS} = 20 \mu\text{A}$	$V_{B1} = 1.2 \text{ V}$
$R_S = 100 \text{ k}\Omega$	
$M_1 M_2 M_3 M_4 M_9$ $M_{10} M_{11}$	$W = 1 \mu\text{m}, L = 0.3 \mu\text{m}$
$M_5 M_6$	$W = 1 \mu\text{m}, L = 0.3 \mu\text{m}$
$M_7 M_8$	$W = 2 \mu\text{m}, L = 0.3 \mu\text{m}$

NMOS: $|V_{thn}| = 0.4 \text{ V}$; $\mu_n C_{ox} = 300 \mu\text{A/V}^2$; $\lambda_n = 0 \text{ V}^{-1}$
 PMOS: $|V_{thp}| = 0.5 \text{ V}$; $\mu_p C_{ox} = 100 \mu\text{A/V}^2$; $\lambda_p = 0 \text{ V}^{-1}$

Differential gain



Circuit parameters

$V_{dd} = 1.8 \text{ V}$	$V_B = 0.9 \text{ V}$
$I_{BIAS} = 20 \mu\text{A}$	$V_{B1} = 1.2 \text{ V}$
$R_S = 100 \text{ k}\Omega$	
$M_1 M_2 M_3 M_4 M_9$ $M_{10} M_{11}$	$W = 1 \mu\text{m}, L = 0.3 \mu\text{m}$
$M_5 M_6$	$W = 1 \mu\text{m}, L = 0.3 \mu\text{m}$
$M_7 M_8$	$W = 2 \mu\text{m}, L = 0.3 \mu\text{m}$

NMOS: $|V_{thn}| = 0.4 \text{ V}$; $\mu_n C_{ox} = 300 \mu\text{A/V}^2$; $\lambda_n = 0.1 \text{ V}^{-1}$
 PMOS: $|V_{thp}| = 0.5 \text{ V}$; $\mu_p C_{ox} = 100 \mu\text{A/V}^2$; $\lambda_p = 0.1 \text{ V}^{-1}$

Take aways

- Biasing
 - Assume region of operation
 - Compute current/voltages
 - ***Verify the regions of operation!***
- Differential circuits
 - Exploit symmetry
- Re-use known circuits/formulas
 - But be careful with the approximations